

Perturbed Gibbs Samplers for Generating Large-Scale Privacy-Safe Synthetic Health Data

Yubin Park, Joydeep Ghosh
Dept. of Electrical and Computer Engineering
University of Texas at Austin

Mallikarjun Shankar
Computational Sciences and Engineering Division
Oak Ridge National Laboratory

Abstract—This paper introduces a non-parametric data synthesizing algorithm to generate privacy-safe “realistic but not real” synthetic health data. Our goal is to provide a systematic mechanism that guarantees an adequate and controllable level of privacy while substantially improving on the utility of public use data, compared to current practices by CMS, OSHPD and other agencies. The proposed algorithm synthesizes artificial records while preserving the statistical characteristics of the original data to the extent possible. The risk from “database linking attack” is quantified by either an l -diversified or an ϵ -differentially perturbed data generation process. Moreover its algorithmic performance is optimized using Locality-Sensitive Hashing and parallel computation techniques to yield a linear-time algorithm that is suitable for Big Data Health applications. We synthesize a public Medicare claim dataset using the proposed algorithm, and demonstrate multiple data mining applications and statistical analyses using the data. The synthetic dataset delivers results that are substantially identical to those obtained from the original dataset, without revealing the actual records.

I. INTRODUCTION

Synthetic data, generated from a certain random process, can address disclosure limitation issues in public use health data. Many health datasets contain privacy-sensitive and sometimes confidential information such as disease, payment, and treatment records. Revealing such health information is clearly disagreeable to many, and raises severe ethical and fiduciary issues. Instead, when created carefully, synthetic data can provide the required statistical information for various analyses in the healthcare domain without revealing person-specific data or person’s identity. Developing appropriate algorithms to generate synthetic data is critical to meeting the growing need for well-grounded health informatics research.

Using traditional “parametric model-based” synthesizers, however, provides only a partial solution to such objectives [1], as it introduces two open-ended issues, namely model selection problem and unquantifiable privacy-risk. The complexity of a synthetic model limits the answerable range of research questions; for example, a linear model synthetic data cannot address identifying quadratic relationships between covariates. However, most data mining research is based on retrospective analysis, where the research questions are posed *post hoc* and may be determined during the data

exploration process. Thus, this type of synthetic data are not perfectly suited for data mining applications. Moreover, popular privacy metrics such as k -anonymity [2], l -diversity [3], or ϵ -differential privacy [4] cannot be directly applied to such model-based synthesizers [5]. Recent reports on adversarial privacy attacks [6], [7] suggest that rigorous characterization of such risk is critical in data publishing.

These difficulties usually result in “overly complicated” synthetic processes when publishing a privacy-sensitive dataset. As an illustrative example, Centers for Medicare and Medicaid Services (CMS) recently published synthetic Medicare data for 5% of the US population [8] – Data Entrepreneurs’ Synthetic Public Use File (DE-SynPUF). To publish such data, the original data were processed with six different Statistical Disclosure Limitation (SDL) techniques: 1) Variable Reduction (feature selection), 2) Suppression (k -anonymity), 3) Substitution (data swapping), 4) Imputation, 5) Data Perturbation (on event times), and 6) Coarsening (binning). These processes usually require extensive domain-knowledge, hence they are very difficult to automate. Furthermore, this chain of perturbations substantially changes the characteristics of the original data, to the extent that CMS explicitly states “As a result, the DE-SynPUF has very limited inferential research utility... That is, analyses using the DE-SynPUF to draw inferences about Medicare beneficiaries, providers, or the Medicare program will be misleading and often incorrect [8, page 16]”.

Non-parametric synthetic data can be a remedy for the model selection issue. Moreover, if the generative process of such data adheres to a certain state of the art privacy metric, then the risk from synthetic data can be well characterized. However, such non-parametric schemes are rarely adopted in generating synthetic data in practice, due to its computational cost, and non-intuitive connections to privacy metrics. In this paper, we analyze the definition of privacy in the healthcare context, and derive its connection to probabilistic generative processes. We propose a novel and practical algorithm for generating synthetic data. The proposed algorithm utilizes parallel computation and Locality Sensitive Hashing (LSH) techniques [9] to address the computational challenges. This data type is dominant in many health records. Furthermore, for many numeric fields,

binning, also known as histogramming or quantization, can be appropriately applied to transform data into categorical format, supporting the generality of our framework.

II. RELATED WORK

A wide variety of privacy oriented mechanisms have been proposed in the past decade, including differential privacy, k -anonymity and l -diversity [10]. We note that while differential privacy is perhaps the currently most popular approach in the academic community, it has not been adopted by healthcare agencies, who rely on a combination of approaches such as described in [8]. Though for brevity we have focussed on l -diversity and differential privacy, our framework can work with a variety of mechanisms, including combinations, making it more likely to be adopted.

There are two main bodies of related work regarding the technical parts of this paper: “statistical privacy-preserving disclosure” and “non-parametric data synthesizing” methods. We first visit Statistical Disclosure Limitation (SDL) techniques, and then briefly cover Markov Chain Monte Carlo (MCMC) to describe data synthesizing methods.

SDL can be cast into three categories [11]: data-swapping, data recoding, and multiple-imputation. In data-swapping [12], sensitive row values are swapped with other row values. Although this method can protect sensitive information from simple adversarial attacks, data swapping distorts the joint distribution of the original data. Data recoding is also known as feature suppression or aggregation. The masking process is usually based on the “ k -anonymity” principle that ensures any entry in data is indistinguishable with other k entries. Data recoding can be viewed as a lossy coding, and finding the utility-optimal coding is an NP-hard problem. Multiple-imputation (MI) [13] utilizes MCMC algorithms to obtain random samples from posterior distributions. MI typically uses parametrized posterior distributions, thus this approach suffers from the same model-based synthesizer issues: model selection and unquantifiable privacy-risk issues.

MCMC methods generally refer to a class of algorithms that draw samples from non-trivial probability distributions through Markov chain simulations. Some special cases are Metropolis-Hastings (MH) algorithm, Gibbs sampler, Slice sampling, and Multiple-try Metropolis. Among many choices, this paper uses Gibbs sampler for two reasons. First, Gibbs sampler only requires conditional distributions and no other tuning parameters. This property is crucial in our approach, since our goal is to control privacy-risks by “deliberately” perturbing conditional distributions. Note that all these alternatives try to estimate the true joint distribution, while we take a novel approach of deliberately perturbing the sampling process to meet privacy requirements while staying close to the true distribution. Second, unlike MH algorithm, Gibbs sampler does not reject samples, thus Gibbs sampler is usually more efficient than MH algorithm when drawing high-dimensional samples. For this reason, Gibbs sampler is

sometimes viewed as a special case of MH algorithm with no rejection.

III. PRELIMINARIES

The original table and synthetic table are denoted as $\mathcal{D}$ and $\mathcal{S}$, respectively. n and m represent the number of rows and columns in table $\mathcal{D}$, and $\mathbf{x} = (x_1, x_2, \dots, x_m)$ is a row of the table. $\mathbf{x}_{\setminus j}$ is a row with the value of x_j undetermined e.g. $\mathbf{x}_{\setminus 1} = (\cdot, x_2, \dots, x_m)$. Random variables are expressed using capital letters, for example X for a scalar, and $\mathbf{X}$ for a vector. $\Pr(\mathbf{X})$ is a true probability distribution of the original data, and $\Pr_e(\mathbf{X} \mid \mathcal{D})$ represents an empirical probability mass function given data $\mathcal{D}$.

A categorical dataset $\mathcal{D}$ with indistinguishable rows, i.e. without any ID’s, can be completely described by a contingency table. Thus, with enough number of samples, the contingency table can be said as the most precise and non-parametric approximation for the underlying joint distribution. We exchangeably use the notations for the normalized contingency table¹ and the empirical distribution $\Pr_e(\mathbf{X} \mid \mathcal{D})$.

The difference between the empirical and the true distribution decays as the number of data points increase [14]. The empirical distribution approaches the true distribution exponentially fast as N increases. In practice, this empirical distribution can be treated as the underlying true distribution when the size of the data is huge.

Obtaining a full contingency table is usually intractable especially with a large number of attributes. However, we can mimic sampling from a full contingency table using Gibbs sampling. Gibbs sampling is a prevalent estimation or sampling technique, when sampling from a joint distribution is intractable. A synthetic sample $\mathbf{x}_s$ from a joint distribution $\Pr_e(\mathbf{X} \mid \mathcal{D})$ can be Gibbs-sampled as follows:

$$\begin{aligned} x_1 &\sim \Pr_e(X_1 \mid \mathbf{x}_{\setminus 1}, \mathcal{D}) \\ x_2 &\sim \Pr_e(X_2 \mid \mathbf{x}_{\setminus 2}, \mathcal{D}) \\ &\vdots \\ x_m &\sim \Pr_e(X_m \mid \mathbf{x}_{\setminus m}, \mathcal{D}) \end{aligned}$$

where $\Pr_e(X_i \mid \mathbf{x}_{\setminus i}, \mathcal{D})$ is a conditional frequency table derived from data $\mathcal{D}$. When the above cycle has reached an enough number iterations (burn-in period), the last sample $\mathbf{x}$ is equivalent to a random sample from the joint distribution. It is important to note that, in Gibbs sampling, the information about the joint distribution is distributed across its conditional distributions.

This brute-force Gibbs synthesizer, however, has three critical issues: a lack of convergence guarantee, computational inefficiency, and loose privacy guarantee. For a Gibbs sampler to converge, the Markov chain in a Gibbs

¹Normalized by the total row counts.

sampler needs to be irreducible and aperiodic. In the brute-force Gibbs synthesizer, the Markov chain of the empirical conditional distributions needs to be verified to satisfy both conditions. Moreover, this brute-force Gibbs synthesizer is computationally expensive. The number of distinct conditional distributions exponentially grows with the number of features, thus pre-computing $\Pr_e(X_i | \mathbf{X}_{\setminus i}, \mathcal{D})$ is not desirable. Estimating $\Pr_e(X_i | \mathbf{X}_{\setminus i}, \mathcal{D})$ on the fly is not a smart choice, since the estimation needs a linear scan for every Gibbs iteration. Finally, synthetic samples from an empirical distribution always can always be found in the original data, as the support of an empirical distribution is the same as the support of the original data. In other words, the synthetic data are “realistic and real”, not “realistic but not real”.

IV. PEGS ALGORITHM

We now present a Perturbed Gibbs Sampler (PeGS). PeGS is illustrated in Algorithm 1. The PeGS algorithm consists of three steps: estimation, perturbation, and sampling steps.

Algorithm 1 PeGS Algorithm

- (Step 1) Estimate $P_e(X_i | g(\mathbf{X}_{\setminus i}), \mathcal{D})$
 - (Step 2) Perturb $P_e \rightarrow Q$
 - (Step 3) Gibbs-sample synthetic data from the perturbed conditional distributions
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A. Step 1: Estimation

Instead of $\Pr_e(X_i | \mathbf{X}_{\setminus i}, \mathcal{D})$, PeGS estimates an approximated distribution, $P_e(X_i | g(\mathbf{X}_{\setminus i}), \mathcal{D})$, where $g(\mathbf{X}_{\setminus i})$ is a hash function such as MinHashing [15] and Locality-Sensitive Hashing (LSH) [9], [16]. The hash function is employed to reduce the number of distinct conditional distributions and to access the conditional distributions faster.

We perform a fast (approximate) nearest neighbor search using LSH. In this paper, we choose to use an LSH family $\mathcal{F}$ defined for a metric space $\mathcal{M} = (\mathbf{x}, \text{Hamming})$, but we note that other distance metrics are also applicable. Thus, even for a really large-scale database with high-dimensional features, PeGS can sample from its (approximated) conditional distributions. Note that there is a trade-off between the exact conditional distribution and the required memory size.

B. Step 2: Perturbation

Two types of perturbation methods are illustrated: l -diversity and ϵ -differential privacy. Although we borrow the privacy metric names, we note that the actual definitions in PeGS are slightly different from the original definitions. For example, the original l -diversity is typically achieved by suppressing or generalizing features, not synthesizing samples. Moreover, the original ϵ -differential privacy is

based on the context of statistical databases, not publishing synthetic data.

We first describe an l -diversified perturbation technique. The obtained conditional distributions are perturbed to meet a prescribed entropy l -diversity criterion for privacy:

Definition 1 (Entropy l -diversity [3]). A probability mass function Q is “synthetic l -diverse” if

$$-E[\log Q] = -\sum_x Q(x) \log_2 Q(x) \geq C \cdot \log_2 l$$

where $0 < \log_2 l < 1$ and $C = \log_2 |\mathcal{X}|$.

We perturb the approximated probability distribution $P_e(X_i | g(\mathbf{X}_{\setminus i}), \mathcal{D})$ to satisfy the l -diversity principle as follows:

$$Q_\alpha(X_i | g(\mathbf{X}_{\setminus i})) = (1 - \alpha)P_e(X_i | g(\mathbf{X}_{\setminus i}), \mathcal{D}) + \alpha U$$

where U is a uniform distribution over the range of X_i . As we want to keep the statistical properties of the original data to the extent possible, we choose the minimum α that satisfies the condition:

$$\alpha^* = \arg \min_{\alpha} Q_\alpha(X_i | g(\mathbf{X}_{\setminus i})) \geq C \log l$$

$$Q(X_i | g(\mathbf{X}_{\setminus i})) = Q_{\alpha^*}(X_i | g(\mathbf{X}_{\setminus i}))$$

Next, we show an ϵ -differential perturbation technique.

Definition 2 (Exponential Gibbs Sampler). A probability mass function Q is “ ϵ -differentially perturbed” if

$$Q_\epsilon(x) \propto \exp(\epsilon \cdot N \cdot P(x))$$

where $\epsilon > 0$ and N is a number of samples that are used to estimate $P(x)$.

In the ϵ -differential perturbation mechanism, a smaller value of ϵ means a larger amount of perturbation. $Q_\epsilon(x)$ approaches to $P(x)$ as ϵ moves toward infinity. In PeGS, this synthetic ϵ -differentially perturbation mechanism can be implemented as follows:

$$Q_\epsilon(x_i | g(\mathbf{X}_{\setminus i})) = \frac{\exp(\epsilon N P_e(x_i | g(\mathbf{X}_{\setminus i}), \mathcal{D}))}{\sum_{z_i} \exp(\epsilon N P_e(z_i | g(\mathbf{X}_{\setminus i}), \mathcal{D}))}$$

This perturbation mechanism guarantees ϵ -differential privacy in a following sense. For two data sets differing at most one row, $\mathcal{D}$ and $\mathcal{D}'$, we have:

$$\sup_{\mathcal{D}, \mathcal{D}'} \frac{Q_\epsilon(x_i | g(\mathbf{X}_{\setminus i}), \mathcal{D})}{Q_\epsilon(x_i | g(\mathbf{X}_{\setminus i}), \mathcal{D}')} < \exp(\epsilon)$$

C. Step 3: Sampling

PeGS samples are generated through:

$$\begin{aligned} x_1 &\sim Q(X_1 | g(\mathbf{x}_{\setminus 1})) \\ x_2 &\sim Q(X_2 | g(\mathbf{x}_{\setminus 2})) \\ &\vdots \\ x_m &\sim Q(X_m | g(\mathbf{x}_{\setminus m})) \end{aligned}$$

This Markov chain converges to a unique stationary distribution, $Q(\mathbf{X})$ if $\alpha > 0$, see Theorem 1.

Theorem 1 (Existence of $Q(\mathbf{X})$). *If $Q(X_i|\mathbf{X}_{\setminus i}) > 0$ (positivity condition), then the Markov chain of a perturbed Gibbs sampler is irreducible, thus there exists a unique stationary distribution $Q(\mathbf{X})$.*

Proof: The Markov chain is recurrent with positivity condition, thus it has a unique stationary distribution $Q(\mathbf{X})$. ■

The difference between $\Pr_e(\mathbf{X})$ and $Q(\mathbf{X})$ is upper-bounded by the parameter $\log l$ and ϵ depending on perturbation methods, and hence the difference between the true distribution $\Pr(\mathbf{X})$ and the PeGS-distribution $Q(\mathbf{X})$ is also upper-bounded. Note that the perturbed Gibbs sampler generates artificial samples that are not in the original data. For these artificial samples, we find the nearest neighboring perturbed conditional distribution as follows:

$$Q(X_i | g(\mathbf{X}_{\setminus i}^{ann})) \quad \text{s.t.} \quad \mathbf{X}_{\setminus i}^{ann} \approx \mathbf{X}_{\setminus i}$$

using LSH, then use this approximated nearest neighbor distribution in the Gibbs sampling iterations.

D. Additional Step: Parallelization

The PeGS Algorithm is embarrassingly parallelizable with sub-sampling. Algorithm 2 illustrates the para-PeGS algorithm. If each process in para-PeGS satisfies the desired privacy definitions, the collection of synthetic data will also meet the desired level of privacy guarantees.

Algorithm 2 para-PeGS Algorithm

- (Step 0) Partition $\mathcal{D}$ into $\{\mathcal{D}^p\}$
 - (Step 1, 2, 3) PeGS with $\mathcal{D}^p$
 - (Step 4) Combine and mix synthetic data $\{\mathcal{S}^p\}$
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V. PRIVACY ANALYSES

The “realistic but not real” principle can be achieved through rigorous analyses on the definition of “privacy” in the healthcare context. Unlike other scientific measurement datasets, healthcare datasets in general contain the information about real people, such as patients and physicians. Thus, the definition of data privacy in healthcare datasets can be narrowed down. We define the goal of privacy protection in the healthcare domain is to protect the identity of the people, and to protect their sensitive information.

The attacks on published datasets are either **identity attacks** (identifying “who”) or **feature attacks** (identifying “what”). Although many other types of attacks may be threats to uncover private information, such attacks are not directly related to our definition of privacy protection. For example, an attacker can link two datasets and find out new information for a specific row, without knowing a person’s identity. In this case, as the attacker has no clue about the

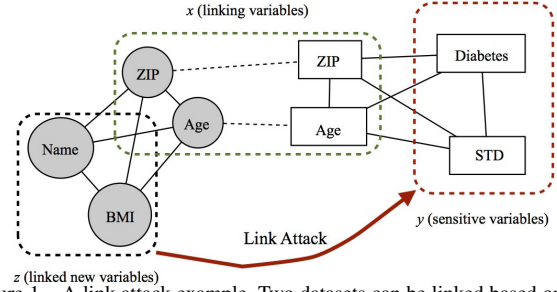


Figure 1. A link attack example. Two datasets can be linked based on Age and ZIP code. Sensitive values such as Diabetes and STD can be revealed using the new linked variables Name and BMI.

identity of the records, this attack does not infringe on privacy by our definition. Again, it is possible that even such information can be useful for later identity and feature attacks (transitive attacks). We note that the uncertainty level of such information may decrease as more linkable datasets become available.

In this paper, we analyze the effect of database linking on feature attacks. Consider a dataset with two features: X (non-sensitive) and Y (sensitive). When publishing the dataset, a typical strategy is to noise Y to be $\tilde{Y}$. However, if another dataset is linked with the original dataset based on X , the linked information may reduce the uncertainty about the sensitive field. Figure 1 illustrates this linking scenario. Theorem 2 illustrates this information gain for attackers by linking two datasets:

Theorem 2 (Link Gain). *Suppose that two datasets $\mathcal{D}_Z = \{Z, X\}$ and $\mathcal{D}_Y = \{\tilde{Y}, X\}$ are linked based on X , then:*

$$\frac{\Pr(\tilde{Y} = Y | X, Z)}{\Pr(\tilde{Y} = Y | X)} = \frac{\Pr(Z | X, \tilde{Y} = Y)}{\Pr(Z | X)} = \Lambda \quad (1)$$

where Λ is the “link gain ratio” by linking two datasets.

Proof: Equation (1) is a direct result by applying Bayes’ theorem. ■

Theorem 2 states two competing objectives when publishing data: Λ needs to be low and $\Pr(\tilde{Y} = Y | X)$ needs to be low. As can be seen, Λ increases as $\Pr(\tilde{Y} = Y | X)$ decreases, and $\Pr(\tilde{Y} = Y | X)$ increases as Λ decreases. Furthermore, Λ can be arbitrarily large when $\Pr(Z | X) \approx 0$, where we have no control over Z and $\mathcal{D}_Z$. Thus, lowering Λ is very difficult. PeGS addresses this issue by perturbing every variable in a dataset i.e. including linking variables. In other words, instead of directly lowering Λ , we inject uncertainty on linking.

The perturbation step in PeGS can be interpreted from the information theory context. The mutual information between X_i and $\mathbf{X}_{\setminus i}$ is as follows:

$$I(X_i; \mathbf{X}_{\setminus i}) = H(X_i) - H(X_i | \mathbf{X}_{\setminus i}) \quad (2)$$

where $I(X_i; \mathbf{X}_{\setminus i})$ is Shannon Mutual Information between X_i and $\mathbf{X}_{\setminus i}$. Uniformly smoothing $\Pr_e(X_i | \mathbf{X}_{\setminus i}, \mathcal{D})$ increases the conditional entropy $H(X_i | \mathbf{X}_{\setminus i})$, decreasing $I(X_i; \mathbf{X}_{\setminus i})$ as a result. Thus, PeGS weakens the link between X_i and $\mathbf{X}_{\setminus i}$ by increasing the conditional entropy. On the other hand, traditional data publishing methods such as k -anonymity and l -diversity decreases the entropy of $H(X_i)$ by generalization or suppression of the feature values. Although their approaches are different, the common goal of such privacy mechanisms is reducing the mutual information between features.

VI. EMPIRICAL STUDIES

In this section, we demonstrate the impact of PeGS algorithm using Medicare Claims records for both descriptive analysis and predictive modeling. Centers for Medicare and Medicaid Services (CMS) provides several public-use data files (PUF), such as inpatient claims, line-items, drug events, etc. For our experiment, we use “BSA Inpatient Claims PUF”² describing Basic Stand Alone Inpatient records. The data file contains seven variables: ID, Gender, Age, DRG (drug code), ICD-9 (procedure code), Length (the length of stay), and Amount (payment), and has 15K rows. Age, Length, and Amount variables are originally numeric records, but CMS has categorized them into five quantiles. Note that data re-coding methods such as k -anonymity and l -diversity cannot be directly compared to PeGS, as their feature granularities are different. To the best of our knowledge, *PeGS is the first privacy-safe data synthesizer, which adheres to the rigorous privacy metrics, “synthetic l -diversity” and “ ϵ -differential perturbation”.*

A. Marginal Distribution

Figure 2 and 3 shows marginal histograms of drug and ICD-9 procedure codes over different levels of privacy metrics $\log l$ and ϵ . Histograms from the original data are shown in the top left cells. As can be seen, the histograms with lower values of $\log l$ and higher values of ϵ are similar to those of the original data. Note that, in an l -diversified PeGS, $\log l = 0$ represents no perturbation on conditional distributions. Also, an ϵ -differential PeGS becomes a non-perturbed PeGS as ϵ approaches to infinity. As can be seen, the histograms when $\log l = 0.5$ and $\epsilon = 0.5$ are very close to a uniform distribution. With this amount of perturbation, any particular record is not distinguishable from other records (privacy-safe), but the utility of the data also decreases. The parameters $\log l$ and ϵ should be determined considering this trade-off relationship between “utility and privacy”.

B. Effect on Predictive Models

Privacy preserving synthetic data can be publicly disclosed to answer various types of data mining research

²http://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/BSAPUFS/Inpatient_Claims.html

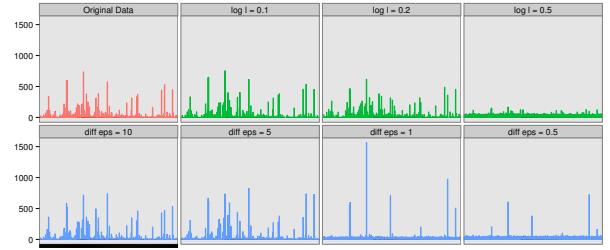


Figure 2. Marginal Histogram of Drug Codes vs. $\log l$ and ϵ .

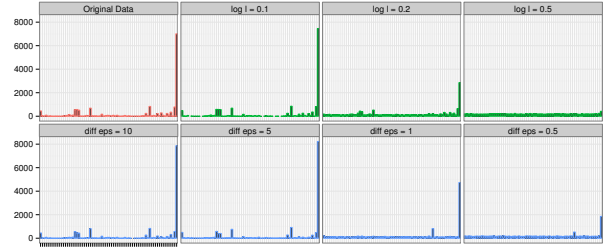


Figure 3. Marginal Histogram of ICD-9 Procedure Codes vs. $\log l$ and ϵ .

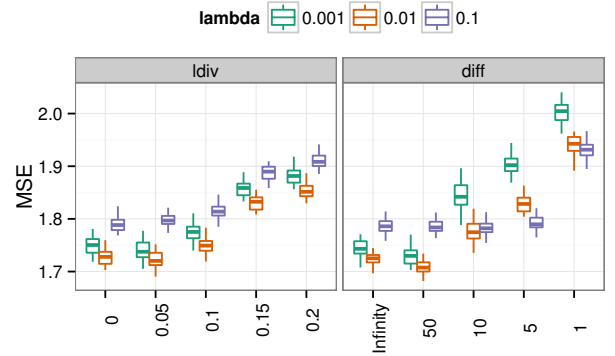


Figure 4. MSE vs. $\log l$, ϵ and λ (color coded, regularization parameter). “ldiv” and “diff” represent l -diversity and ϵ -differential privacy, respectively.

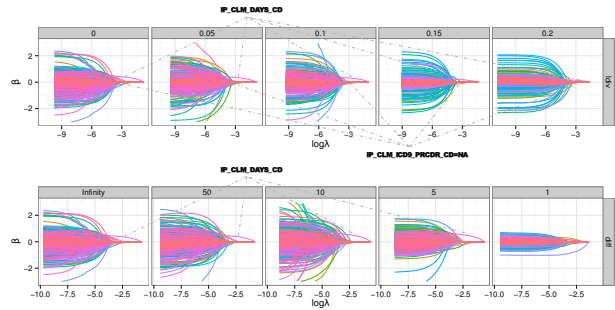


Figure 5. Coefficient values vs. $\log l$, ϵ and λ . Each β is color-coded. “IP CLM DAYS CD” (days) is positively correlated with the target variable. “IP CLM ICD9 PRCDR CD=NA” is negatively correlated with the target, and we also can check this fact from the l -diversified PeGS data.

questions. If the statistical properties of the original data are well preserved in the PeGS-generated data, then the data mining results from the synthetic data should be significantly identical to those from the original data. We demonstrate illustrative results of simple predictive modeling by arbitrarily setting Amount as a dependent variable:

$$\underbrace{\text{AMT}}_{y: \text{dependent variable}} \sim \underbrace{\text{Age} + \text{Gender} + \text{DRG} + \text{ICD9} + \text{DAYS}}_{X: \text{independent variables}}$$

Dummy coding is applied to the categorical variables such as DRG and ICD9, resulting in 352 dependent variables. We apply the Lasso regression [17] to deal with this high dimensionality, then measure Mean Square Errors (MSE) using the original data:

$$\beta^* = \min_{\beta} \|\mathbf{y}_{\text{synth}} - \mathbf{X}_{\text{synth}}\beta\|^2 + \lambda\|\beta\|$$

$$\text{MSE} = \|\mathbf{y}_{\text{orig}} - \mathbf{X}_{\text{orig}}\beta^*\|^2$$

Figure 4 shows the measured MSE values over different λ , $\log l$, and ϵ values. The regularization parameter λ is color-coded. As can be seen, MSE values increase with the $\log l$ value, but their deviation from the original MSE is negligible compared to the spread of values at each privacy setting, $\log l$. For ϵ -differentially perturbed data, we can observe that the measured MSE values decreases as ϵ approaches to infinity.

Next we check the interpretation of the models obtained from the perturbed data. Figure 5 shows the coefficients values β^* over $\log l$, ϵ , and λ values. Intuitively, the amount spent in a hospital (the target variable) is proportional to the days spent in a hospital. From the original data, we can observe a strong correlation between these two variables. As a result, LASSO regression produces a non-zero coefficient value for this independent variable (days) when a strong regularization parameter is imposed (e.g. $\log \lambda > -3$). As can be seen in Figure 5, the perturbed data produce non-zero coefficients for the “day” variable. Although we also can see some differences in other non-significant coefficient values, the perturbed data preserve the overall coefficient characteristics of the original data.

VII. DISCUSSION

We proposed a novel data synthesizer that protects sensitive information by adhering to a prescribed l -diversity and ϵ -differential privacy metrics. The proposed algorithm, PeGS, scales linearly with respect to the size of the data. Furthermore, PeGS is a non-parametric and non-model based technique. This property assures that a wide range of data mining algorithms can be applied without considering the generative process of the synthetic data. Many health datasets are not available because of privacy restrictions such as HIPAA privacy regulation. We believe that the proposed solution can be an alternative way to facilitate collaborative healthcare research efforts.

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